Land Use Policy 113 (2022) 105932

Contents lists available at ScienceDirect

Land Use Policy

journal homepage: www.elsevier.com/locate/landusepol

Impact of protected areas and co-management on forest cover: A case study from Teknaf Wildlife Sanctuary, Bangladesh

S M Asik Ullah [a,*], Masakazu Tani [a], Jun Tsuchiya [a], M.Abiar Rahman [b], Masao Moriyama [c]

[a] Department of Environmental Design, Kyushu University, Japan
[b] Department of Agroforestry and Environment, BSMR Agricultural University, Bangladesh
[c] Faculty of Engineering, Nagasaki University, Japan

ARTICLE INFO

Keywords:
Protected area
Co-management
Deforestation
Local community

ABSTRACT

Deforestation is a matter of serious global concern. Establishing protected areas (PAs), which cover nearly 15% of the world's surface, is one of the leading approaches to forest conservation. The recent global expansion of PAs has led to the inclusion of local communities in the forest management process, an approach referred to as "co-management." PAs under co-management are expected to conserve biodiversity while providing support to local communities. This trend for forest co-management has shown mixed results, and a better understanding of its impact on forests and people is required. This study attempts to explore the forest-cover change in a PA of Bangladesh—the Teknaf Wildlife Sanctuary (TWS). Remote sensing will be used to assess the land-cover change, followed by a household survey to define the impact of local settlements responsible for this change. To analyze the impact of PA and co-management on forest cover, satellite images from 1989, 2007, 2009, and 2015 were considered due to the availability of quality images. The study found that from 1989 to 2015, the region saw a 46% decrease in forest cover compared to 64% inside the PA boundary; deforestation occurred more inside the PA than outside. Moreover, after the implementation of a co-management approach, the rate of deforestation increased inside the PA. Besides the forest conversion, the tree coverage shifted from the core forest area toward the settlement area, causing serious negative impacts on biodiversity. The household survey listed 57,404 households in the region, among which 5195 were situated inside the PA. Co-management in PAs was intended to involve local communities in forest management, but this study showed that the local community is causing more deforestation and shifting the tree coverage towards the forest boundaries. The creation of a large, multi-use buffer zone to ensure the long-term tenure of the core forest area could be a viable option. Such buffer zones can provide effective security and enhance local people's livelihoods.

1. Introduction

Forest loss has been occurring for a long time. However, the recent rapid rates of tropical deforestation—nearly 6 million ha of forests per year between 2010 and 2015—have raised serious concern (FAO, 2016). Preventing or, at a minimum, slowing deforestation on a global and national scale represents an enormous challenge and requires specific land-use policies addressing forest loss. Land-use policies and forest conservation strategies to avoid deforestation and mitigate the consequences are contentious because the drivers of deforestation differ at temporal and spatial scales (Geist and Lambin, 2002; Pfaff et al., 2013; Rudel et al., 2009). Establishing protected areas (PAs) is the most commonly practiced land-use policy to protect forests and conserve biodiversity (Watson et al., 2014a, 2014b). This land-use policy is not new, but, over the past century, the concept of PAs has been redefined, developed, and modernized as the number and coverage multiplied several folds worldwide—from a handful of sites at the turn of the 20th century to 258,608 designated PAs covering 20.28 million square kilometers in 2020, equivalent to 15% of the earth's land surface (UNEP-WCMC, 2020; Watson et al., 2014a, 2014b). Undoubtedly, PAs have conserved biodiversity better than any other land-use policy (Butchart et al., 2012; Geldmann et al., 2013a; Joppa and Pfaff, 2010; Naughton-Treves et al., 2005a). However, the rapid expansion of both PAs and human settlements from the latter half of the 20th century has increased their contact with each other (Watson et al., 2014a, 2014b). Previously established PAs in remote areas are now facing new

* Correspondence to: Department of Environmental Design, Kyushu University, 4-9-1 Shiobaru Minami Ku, Fukuoka 815-8540, Japan.
E-mail address: ullah.asik.s.m.817@m.kyushu-u.ac.jp (S.M.A. Ullah).

https://doi.org/10.1016/j.landusepol.2021.105932
Received 25 October 2020; Received in revised form 6 December 2021; Accepted 8 December 2021
Available online 15 December 2021

challenges because substantial communities have been developed near to them, and issues related to the exclusion of local communities from PAs have become a serious concern (Agrawal and Redford, 2009; Brockington et al., 2006).

In response, during the late 70 s and the early 80 s, the concept of including local communities in PA management began to gather momentum (Lemos and Agrawal, 2006). During the 80 s and later on, extensive research by scholars on common property, political ecology, community capacity, and resource management by other small-scale social formations provided the intellectual basis for a shift toward decentralized environmental policies and community-based resource management (Lemos and Agrawal, 2006; Elinor Ostrom, 1990a, 1990b; Peluso, 2012). Now, local communities legally manage nearly 15.3% of the world's forests, and over half a billion people depend on community-managed forests (Agrawal, 2007; RRI, 2017). Such forest management approaches are described under various terms in different countries; examples include community forestry, community forestry user group, social forestry, co-management, and community-based conservation (FAO, 2020). "Co-management" is used in this paper as a generic term to cover the above-mentioned approaches in the form of different projects and programs in different countries and is defined as the management of reserved and protected forests by the government, where surrounding communities are actively involved in designing, planning, and executing policy (Matose, 2006).

Over the last few decades, governments and international organizations have promoted the geographical and conceptual expansion of co-management for the governance of PAs, particularly among the developing countries in the tropical region. Now, PAs are created to both conserve the forest alongside its habitat and provide livelihood opportunities for the local communities. This additional role of PAs has promoted justification for their expansion; however, on top of their already questionable effectiveness, it has made the PAs more likely to fail in meeting their added objectives (Geldmann et al., 2013a; Muench and Martínez-Ramos, 2016; Naughton-Treves et al., 2005b). The effectiveness of PA co-management is mainly undermined by a lack of resources, especially in underdeveloped and developing countries (Balmford et al., 2003; Bruner et al., 2004; Leverington et al., 2010). The rapid expansion of PAs has outpaced their monitoring and evaluation and has created a void in understanding the impact and effectiveness of co-managed PAs in conserving biodiversity. Therefore, to protect the forests, an in-depth understanding of the outcomes of these PAs on global, national, and regional scales is necessary.

Bangladesh is a developing country with only 11% of its total land under forest (FAO, 2015), and the surviving forests are losing tree cover at an alarming rate (Potapov et al., 2017). All natural forests in Bangladesh are state owned and managed by the Bangladesh Forest Department (BFD). Co-management of PAs was introduced in Bangladesh in 2004, comparatively later than neighboring countries, including Nepal and India (BFD, 2020). Recently, co-management approaches have gained popularity with local people who depend on forests and also showed some sustainable positive impact on biodiversity conservation (Mukul et al., 2012; Rashid et al., 2013b). On the other hand, there are serious concerns regarding these co-managed PAs, including the effectiveness of the system to conserve forests, lack of active participation from the local communities, ambiguity regarding the roles and responsibilities of the stakeholders, and an implementation gap in policy frameworks (Mollick et al., 2018; Rahman et al., 2017; Rashid et al., 2013a). The establishment of new PAs (aiming to cover 30% of state-owned forests by 2035) and the expansion of co-management approaches to govern and manage these PAs are firmly embedded in the forestry policies of Bangladesh (BFD, 2016). However, the existing policy and legal frameworks are not yet adequate to support such expansion initiatives, a fact that has become abundantly clear through the enactment and frequent changes to forest policies, rules, and acts within the last decade (Rahman et al., 2016b). The BFD is upscaling the existing model of co-management to more PAs, with very few adaptations, through aid-funded projects from international donor agencies (Rahman et al., 2016a). Because there is no "one-size-fits-all" solution, copying an apparently successful model of co-management may not work for all the PAs in Bangladesh.

While the rapid expansion of PAs continues globally, a shift from quantity to quality is essential to maintain their effectiveness (Pringle, 2017; Watson et al., 2016). To improve the quality and effectiveness of PAs in future, it is important to determine the impact and outcomes of the existing ones. There are a handful of studies regarding the impacts and outcome of PAs conserving biodiversity, taking into consideration large numbers and covering global scales (Hajjar et al., 2020; Oldekop et al., 2019a; Watson et al., 2014a, 2014b; Yang et al., 2021). Besides these global- and national-scale studies, in-depth micro-level studies are important as well because they can provide a detailed understanding of the impact of PAs at local and regional scales. The findings of such research could be significant in reducing the gap between management plans and practical realities while governing the PAs. This study aims to contribute by exploring the impact of co-management approaches on a PA referred to as the Teknaf Wildlife Sanctuary in the southern coastal region of Bangladesh, where continued deforestation has been occurring for years (Islam et al., 2019a, 2019b, 2019c; Moslehuddin et al., 2018). Teknaf Wildlife Sanctuary (TWS) is one of the five PAs in Bangladesh where co-management approaches were initiated in 2004. The findings from this empirical research will generate advanced understanding and knowledge essential to land-use policy formulation and future forest conservation in Bangladesh. Thus, the focus of this study is to (1) explore the land-cover changes in the Teknaf region from 1989 to 2015, (2) compare the land-cover changes between inside and outside the TWS, and (3) determine the impact of settlements and the co-management approach on the forest-coverage changes.

Previous studies have explored the forest-cover change in numerous PAs in Bangladesh (Billah et al., 2021; Rahman and Islam, 2021; Zafar et al., 2021), including several concerning PAs under the co-management approach (Ahmed and Bartlett, 2018; Chowdhury et al., 2020; Islam et al., 2020, 2019a, 2019b, 2019c), and most of them reported deforestation at various rates. This research, however, considered several spatial parameters while exploring the land-cover change, helping to provide a critical understanding of the forest conservation process in PAs. Comparisons between the forest-cover change in the core forest area and the surrounding community gave a better perspective of the role of the local community on the deforestation process. The land cover change between inside and outside the TWS was also compared, which helped to illuminate the impact of PA establishment on forest-cover conservation. To our knowledge, this type of land-cover change analysis considering various spatial parameters is among the first studies of its type concerning PAs in Bangladesh. This critical understanding of deforestation of PAs and the impacts of the surrounding community will be helpful for forest managers and policy-makers to protect the forests from further destruction and develop sustainable land-use policies for the remaining forests.

2. PA governing policy and theoretical context

2.1. Protected areas and their managing policies in Bangladesh

PAs are declared, governed, and managed by different legislative tools (e.g., forest policies, acts, and rules), which are enacted and amended by the government of Bangladesh (GoB) where the BFD has the sole governing authority. Currently, there are 60 PAs in Bangladesh, among which 23 have been established in the last decade (BFD, 2021). Fig. 1 presents important forest legislation with the number of PAs in the respective timeframe. The first legislative initiative from GoB was the Wildlife (Preservation) Act of 1974, where the national responsibility for conservation was articulated, and the PAs were categorized as national parks, wildlife sanctuaries, and game reserves, corresponding to IUCN categories II, IV, and VI, respectively (Chowdhury and Koike, 2010).

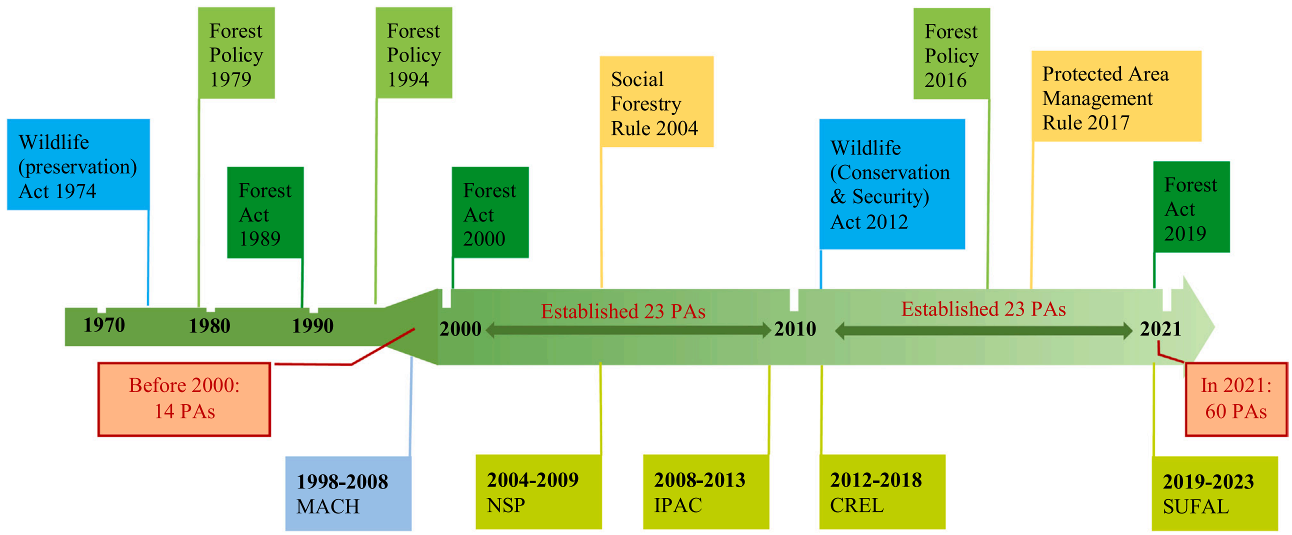

Fig. 1. Timeline of forest policy and PA establishment in Bangladesh.

During the subsequent four decades, a handful of policy and legislative efforts were taken to manage and protect the forests. The Forest Act, 1927 is the backbone of the forestry legislative efforts of the GoB and was adopted from the Indian Forest Act 1927 and amended in 1989, 2000, and 2019 to accommodate new forest management approaches. Besides the Forest Act amendments, three forest policies were developed and decreed in 1979, 1994, and 2016.

The 1979 policy treated the PAs as a location for recreational hunting by calling for "utilizing the recreational potentials of forests." However, later policies considered PAs as conservation tools; the 1994 policy stated that PAs should expand to 10% of the total forest area by 2015, and the 2016 policy planned to expand PAs to 30% by 2035 (BFD, 2016, 1994). In 2004, the Social Forestry Rule was enacted and later amended in 2010 and 2011. This rule provided legislative directives that included local communities in PA management by permitting the planting of tree plots inside and around PAs (BFD, 2004). The Social Forestry Rule had a significant impact on increasing the tree coverage across the country but faced criticism due to the planting of invasive tree species in natural forests (Muhammed et al., 2008).

The Wildlife (Conservation and Security) Act, 2012, was enacted with a special focus on governing the PAs. This act included important provisions regarding the declaration of PAs, their management plans and procedures, prohibited activities inside the PAs, and co-management approaches (BFD, 2012). This act replaced all "game reserve" protected areas with "wildlife sanctuaries," which reflected the legislative efforts to strictly prohibit any recreational hunting inside the PAs. Continuing these efforts, the Protected Area Management Rule, 2017, was enacted on November 12, 2017. It was a specific set of guidelines regarding how to manage the PAs in Bangladesh. This rule defined clear procedures for the formation of co-management committees (CMCs) at different levels to conserve the PAs, specified who should be the main beneficiaries of the forest resources, set out penalties for illegal encroachment, and included details about the distribution of forest resources (BFD, 2017). However, there are pushes for more specification and clarity in the legislative efforts to achieve the desired conservation output from PAs.

2.2. Co-management approaches for PAs in Bangladesh

Forest policies in Bangladesh are highly influenced by foreign donor agencies, including USAID, UNDP, the EU, GIZ, World Bank, and ADB. These donor agencies initiate and fund the development and implementation of forest policies in the form of development project aid (DPA). With the funds from DPA (either loan or aid), the donor agencies partner with the BFD (along with other related GOs and NGOs) to develop and implement forest-related projects. Donors also advise and support the BFD in developing and amending particular forest policies under these projects. The majority of the forest policy changes, the development of new policies in the last few decades, and the role of the donor agencies in these policy initiatives have been identified and described in several published articles (Rahman et al., 2016b, 2016a; Sadath and Krott, 2012). From 1980–2014, community-based forestry received 350.86 million USD as DPA and accounted for 17 major policy changes (Rahman et al., 2016a). The cash flow from donor agencies increased significantly in the early 2000s, resulting in a higher number of PA declarations and more policy enactments and amendments concerning community engagement, including the Social Forestry Rule, 2004; Wildlife (Conservation and Security) Act, 2012; Protected Area Management Rule, 2017; Forest Policy, 2016; and Forest Act, 2019 (Fig. 1).

Consequently, among the PAs, 22 are currently under the co-management approach (Thompson et al., 2018). This approach was initiated in Bangladesh through five major DPA-funded projects, namely Management of Aquatic Ecosystems through Community Husbandry (MACH), Nishorgo Support Project (NSP), Integrated Protected Area Co-management (IPAC), Climate Resilient Ecosystems and Livelihoods (CREL), and Sustainable Forests and Livelihoods (SUFAL). The projects were supported mainly by USAID and World Bank, namely (duration, fund in million USD) MACH (1998–2008, 14), NSP (2004–2009, 9), IPAC (2008–2013, 13), CREL (2012–2018, 36), and SUFAL (2019–2023, 175) (Aziz and Decosee, 2009; Mackenzie et al., 2013; Thompson and Chowdhury, 2007; Thompson et al., 2018; World Bank, 2018). MACH was the first to implement co-management approaches in Bangladesh and introduced community-based organizations (CBOs) formed with the local communities, which were then formally linked with the local government to manage the common natural resources (Thompson and Chowdhury, 2007).

NSP (2004–2009) was the first project to implement a co-management approach for managing PAs in the forestry sector. The aim of NSP was to test and develop a functional model of a co-management approach for the terrestrial PAs. NSP covered 5 PAs (nearly one-third of the total Pas during the project implementation period) and formed co-management councils and committees (CMCs). These CMCs were formed from local government officials, BFD officials, and local citizens to act as the core element for implementing forest co-management. The members of the CMCs received various types of training and functioned as the primary authority to manage the forests. Later on, these CMCs were recognized by the government, and CMC formation was embedded in the Wildlife (Conservation and Security) Act, 2012, under chapter IV, section 21 (BFD, 2012).

After NSP, IPAC (2008–2013) significantly scaled up the co-management approach in both forests and wetlands and strengthened the institutional capacity of the co-management organizations (CMOs). IPAC put CBOs and CMCs under the same umbrella, and they were referred to as CMOs. In addition, the CMOs had representatives from village-level organizations termed people's (PFs) and village conservation (VCFs) forums. The CMOs were responsible for governing and managing the forests and sharing the resources with the local community while maintaining sustainability. At the end of the IPAC, there were 55 CMOs, including 1000 village-level organizations covering 25 PAs, of which 22 PAs were forest lands and 20 forest-related policy changes, including declaring 9 new PAs (Mackenzie et al., 2013).

Later on, CREL (2012–2018) worked to make the CMOs sustainable and incorporated ecosystem-based approaches (EbAs) (USAID, 2018). The EbAs encouraged the CMOs to manage and conserve natural resources and share the benefits generated through different ecosystem services. CREL covered 45 CMOs, and 965 villages followed EbAs to conserve natural resources and benefit at the same time (USAID, 2018). From the policy influence perspective of CREL, the Protected Area Management Rule, 2017, was enacted during the project implementation, which recognized the EbA where 50% of the income generated from forests was distributed to the responsible CMOs (BFD, 2017).

The latest DPA-funded project to promote co-management is SUFAL, which is expected to be completed in 2023. SUFAL aims to scale up the co-management approaches to more PAs and strengthen the institutional capacity of the CMOs, with a special focus on the impoverished people who depend on the forest. Finally, the co-management approach in Bangladesh is still now a DPA-funded, project-based venture with collaboration between the BFD and donor agencies. A serious concern remains regarding the future of co-management when the donor agencies stop funding.

2.3. Theoretical context of co-management in PAs

Governing, managing, and conserving natural forests alongside local communities is a complicated task. To understand the evolution and implementation of community-based natural resource management, two concepts—"the tragedy of the commons" and "governing the commons"—must be considered. "The tragedy of the commons" refers to the expected degradation of the environment when people use a scarce resource in common (Hardin, 1968). Here, every individual acts independently and rationally according to their own self-interest and against the interests of the whole community by depleting a common resource. According to this theory, without strict rules and regulations, the common forests will be destroyed due to over-exploitation by the local communities, which is the concept behind a centralized forest governance system. The establishment of PAs is an example of such centralized approach, where local communities are restricted from forests in order to conserve them based on an assumption of the community's lack of ability to manage natural resources. This restriction can be physical (prohibited from entering the forests), economic (banning or limiting extracting forest resources for local livelihood), or both.

In the 1980 s, due to concerns over the effectiveness of PAs in conserving biodiversity and the exclusion of local communities from the forests, the centralized approach of forest governance began to change, with a paradigm shift toward decentralization. Several scholars had questioned the preconceived notion of the inability of local communities to manage common resources. New theories and ideas emerged, stating that local communities could manage natural resources effectively and provide benefits from sustainable resource harvesting under specific arrangements and agreements. Among the theories, the most popular and iconic was proposed by Elinor Ostrom in 1990, referred to as "Governing the Commons." Ostrom (1990a) (1990b) identified 8 principles required for effective common property resource management and stated that, under these principles, "the tragedy of the commons"' can be avoided. There are concerns over the precise number of principles and their relative importance for the co-management of natural forests (Agarwal, 2001; Cox et al., 2010). However, most scholars agree that secure property rights, effective local institutions, and community incentives are prerequisites for successful co-management (Pagdee et al., 2006).

3. Methodology

3.1. Study area

This study focused on a PA referred to as the Teknaf Wildlife Sanctuary (TWS), which was among the first batches of PAs to adopt co-management in Bangladesh. TWS is situated in the southern coastal area under the Teknaf Upazila (Upazila is an administrative unit under a district) of Bangladesh. Teknaf Upazila (20° 51′ 56′ N and 92° 17 43′ E) is located 48 kilometers south of Cox's Bazar District and bordered by the Bay of Bengal from west to south, the Naf River to the east, and Ukhia Upazila to the north (Fig. 2). Because three sides of this Upazila are bordered by waterbodies, it is also known as the Teknaf Peninsula. The dimensions of the TWS are approximately 28 km from north to south and 3–5 km from east to west, with a total area of 11,615 ha. TWS is among the oldest PAs in Bangladesh and has a rich biodiversity. The forested areas in the Teknaf Peninsula first became a reserve forest in 1907, referred to as the Teknaf Game Reserve (Aziz and Decosee, 2009). TWS is among the very few PAs where large mammals roam in the wild. TWS is well known for being the last recognized habitat in Bangladesh for the Asian elephant (*Elephas maximus*) and the large Indian civets (*Viverra zibetha* and *Viverricula indica*), both of which are endangered species (according to an IUCN report in 2004, only 15–100 elephants remain) (Alam et al., 2013). The forest cover in TWS is broadly classified as mixed tropical evergreen and semi-evergreen forest.

Because TWS is situated in a peninsula, the surrounding settlements are located alongside the PA boundary. According to the latest household census (BBS, 2011), the total population of Teknaf Upazila is 264, 389, comprising 46,328 households.

The region has reserved the forest for a long period of time; therefore, there is no presence of heavy industries, large factories, or even large-scale logging facilities (Tani and Rahman, 2018b). Due to a lack of livelihood opportunities and a comparatively higher poverty level than other regions in the country, the local people highly depend on the natural forest and its resources (Moslehuddin et al., 2018).

Even with the lack of large-scale deforestation drivers, anthropogenic pressure from the local community is causing deforestation. Previous studies have reported that fuelwood harvesting, cash crop cultivation by small farmers, and settlement encroachment are among the main deforestation drivers that have been causing continuous forest conversion during the past few decades (S Alam, 2015; Tani, 2018a; Ullah et al., 2020; Ullah and Tsuchiya, 2018). Involving the local community in forest management while they are responsible for most of the recent deforestation is a challenging task and requires a better understanding of the relationship between the forest and people.

3.2. Land cover change by remote sensing

To investigate the land-cover change of forests, Land Remote-Sensing Satellite (Landsat) images are among the most widely used scientific methods by researchers (Geldmann et al., 2013b; Venter et al., 2014). In this study, Landsat images from 1989, 2007, 2009, and 2015 were used because they offered the most visibility and clear pictures of the study area. The considered Landsat images had a spatial pixel resolution of 30 m × 30 m. Landsat image pixels within the study area were processed and categorized according to the normalized difference vegetation index (NDVI) values. NDVI detects land-cover use based on vegetation coverage and is widely used to detect forest-cover change.

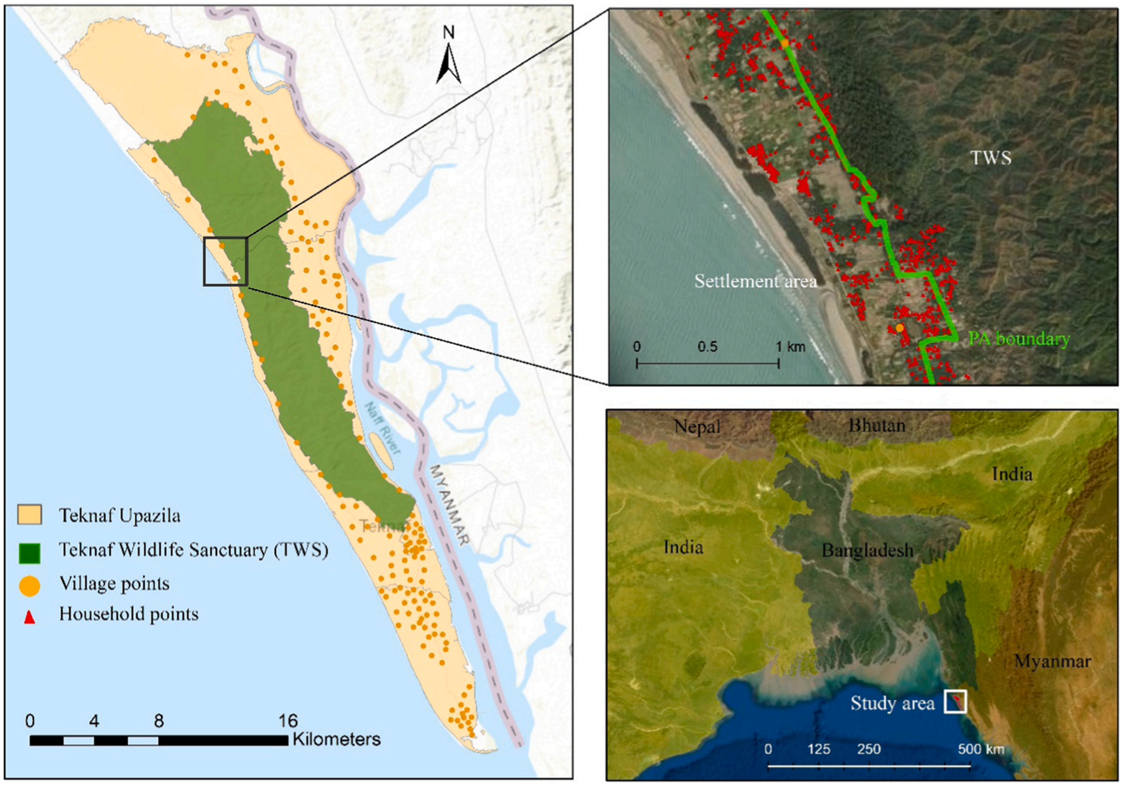

Fig. 2. Study area showing TWS and surrounding households.

Higher NDVI values indicate more greenness and can be related to high tree coverage (Tucker, 1979).

In this study, Landsat pixels with an NDVI value less than 0.40 (NDVI ranges from +1 to −1) were first eliminated from the analysis due to poor vegetation coverage and were termed Class 0 (barren land class). The remaining pixels were classified by changes in NDVI during the dry season of the years 1989, 2007, 2009, and 2015. For this study, pixels were clustered using the iterative self-organizing data analysis technique (ISODATA) based on a series of NDVI values obtained for every month from the beginning of the dry season (November) to its conclusion (April) during the considered years. As a result, three classes of were identified: Class 1, Class 2, and Class 3, of which pixels of Class 3 had the highest NDVI value, and these areas were postulated to contain mature trees or forest. Class 3 was later subjected to further analysis considering ground truth data from 63 locations and three types of spectral characteristics: NDVI, normalized difference water index (NDWI), and green-red vegetation index (GRVI). This analysis yielded three new classes (3, 4, and 5) produced out of Class 3, resulting in a total of five classes. Of the five classes, Class 1 had the lowest NDVI; the NDVI value gradually increased from Class 1–5, and Class 5 had the highest NDVI value.

To define and validate the land-cover classes, the ground-truth method was used. Classes 1 through 5 were compared against ground photos taken by a GPS-enabled GoPro camera during a 3 km transect walk inside the TWS. After initial screening, 515 photos with clear views were selected and categorized into 6 groups based on their vegetation attributes, namely, grassland with small bushes, short bush, tall bush, bush area with low tree coverage (<10% canopy coverage), lean forest (10–40% canopy coverage), and forest (>40% canopy coverage). Sample photos of each group are attached in Appendix 1 and were juxtaposed with the pixels of a 2015 image (because the photos were taken in November 2016, there was justification for comparing them with 2015, the nearest time). Table 1 presents the frequency of the photos in each class. According to the results from Table 1, the classes were defined as Class 1 – Grass, Class 2 – Bush, Class 3 – Mixed, Class 4 – Lean Forest, and Class 5 – Forest.

3.3. Spatial parameters

This study explored different spatial parameters related to socio-economic aspects and forest-cover change. To generate and analyze the spatial parameters, ArcGIS software (version 10.3, ESRI Japan) was employed. Different spatial data attributes were obtained from maps, surveys, satellite images, and other sources and then converted to geographic information system (GIS) layers using ArcGIS. The GIS layers were registered under the UTM 46 N geographic coordination system. TWS refers to the PA area of the Teknaf Peninsula. The boundaries of the

Table 1
Land-cover class comparison with ground photo.

Land cover classes from satellite images	% of ground photo by different categories						Total number of photos
	Grassland with small bushes	*Short bush*	*Tall bushes*	*Bush area with low tree coverage*	*Lean forest*	*Forest*	
Grass	50%	30%	13%	7%	–	–	***24***
Bush	22%	31%	36%	11%	–	–	***250***
Mixed	16%	8%	20%	40%	16%	–	***80***
Lean forest	7%	7%	7%	6%	67%	6%	***65***
Forest	–	–	–	–	3%	97%	***96***

TWS were identified and extracted from the *Mouza* maps and converted to GIS layers. The Land Record and Survey Department (http://www.dlrs.gov.bd) of Bangladesh provides *Mouza* maps that contain the detailed mapping of the settlement areas under different administrative units (Fig. 2). The GIS layer of TWS was used to identify the forest boundaries and determine the encroachment status of the households.

A household survey was conducted from December 2015 to May 2016 that contained location information (latitude and longitude) of all houses in Teknaf Upazila, collected using handheld global positioning (GPS) equipment. The survey covered all 147 villages comprising 57,404 households, 5195 of which were inside the TWS boundary. These household points were converted to GIS layers and used in settlement impact analysis (Figs. 2 and 3). To analyze the local community's impact on the forest cover, a spatial parameter was generated by using the "Buffer" command of ArcGIS to create a buffer area surrounding the households. Based on these household surrounding areas, the study area was divided into settlement and forest zones. Both of these zones are complementary, where the settlement zone is the surrounding area of the households, and the remaining area is the forest zone (Fig. 3). To understand the impact of settlements, 100 m, 200 m, 500 m, and 1 km distances were considered to create buffer areas surrounding the households (Fig. 3). For each distance considered, the settlement zones were termed "settlement 1 km," "settlement 500 m," "settlement 200 m," and "settlement 100 m" for the 1-kilometer, 500-, 200-, and 100-meter areas surrounding the households, respectively. Based on similar buffer diameters, forest zones were termed "forest 1 km," "forest 500 m," "forest 200 m," and "forest 100 m." The total area considered for spatial analysis was 20,577 ha, which excluded beaches, riverbanks, and salt fields. Fig. 3 shows the respective zones with the area coverage, the increase in settlement zones, forest zones reduction, and vice versa.

4. Results

4.1. Land-cover change from 1989 to 2015

The land-cover change according to different land-cover classes are presented in Figs. 4 and 5. The time series considered here were 1989, 2007, 2009, and 2015. From 1989–2015, a total of 1735 ha of forest land cover was lost, and forest decreased from 18% to 10% of the total area. In total, the forest cover lost 46% of its area since 1989. Forest was the only land-cover type to decrease in this period.

Among the increased land-cover classes, bush increased the most, followed by grass, mixed, and lean forest. Bush increased from covering 22% of the total area in 1989–34% in 2015, with a total increase of 2275 ha. Bush was followed by grass, which increased from covering 13% of the total area to 16% of the total area between 1989 and 2015. Mixed and lean forest area increased less compared to other land-cover types; mixed increased by 266 ha, and lean forest increased by only 28 ha.

The land-cover scenario was different in 1989, where bush was the dominant land class (4624 ha) followed by forest (3734 ha). The land-cover transformation process followed a complex pattern, resulting in expanding and shrinking land classes due to various drivers, mostly anthropogenic pressures from the local community. After 1989, the bush and grassland classes constantly expanded, resulting in 49% and 17% increases, respectively. The mixed and lean forest land class did not change significantly, with only a 12% and 1% increase during the time period. However, the transformation of lean forest followed a U-curve pattern with an initial decrease of 30% in 2007, followed by a gradually increasing trend, surpassing the initial coverage by a small margin in 2015. Forest land class was the only land-cover type to decrease almost

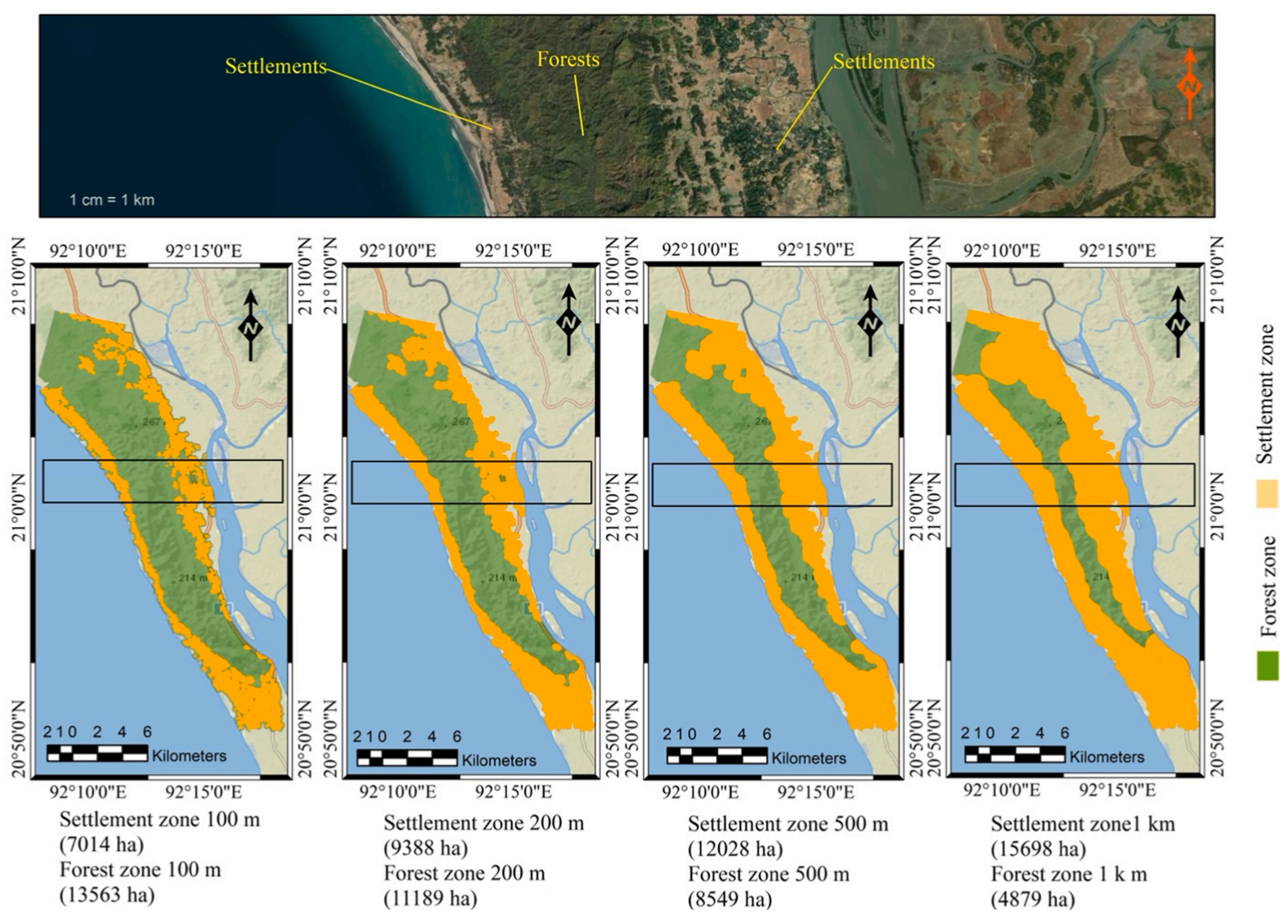

Fig. 3. Forest zones and settlement zones based on different buffer diameters.

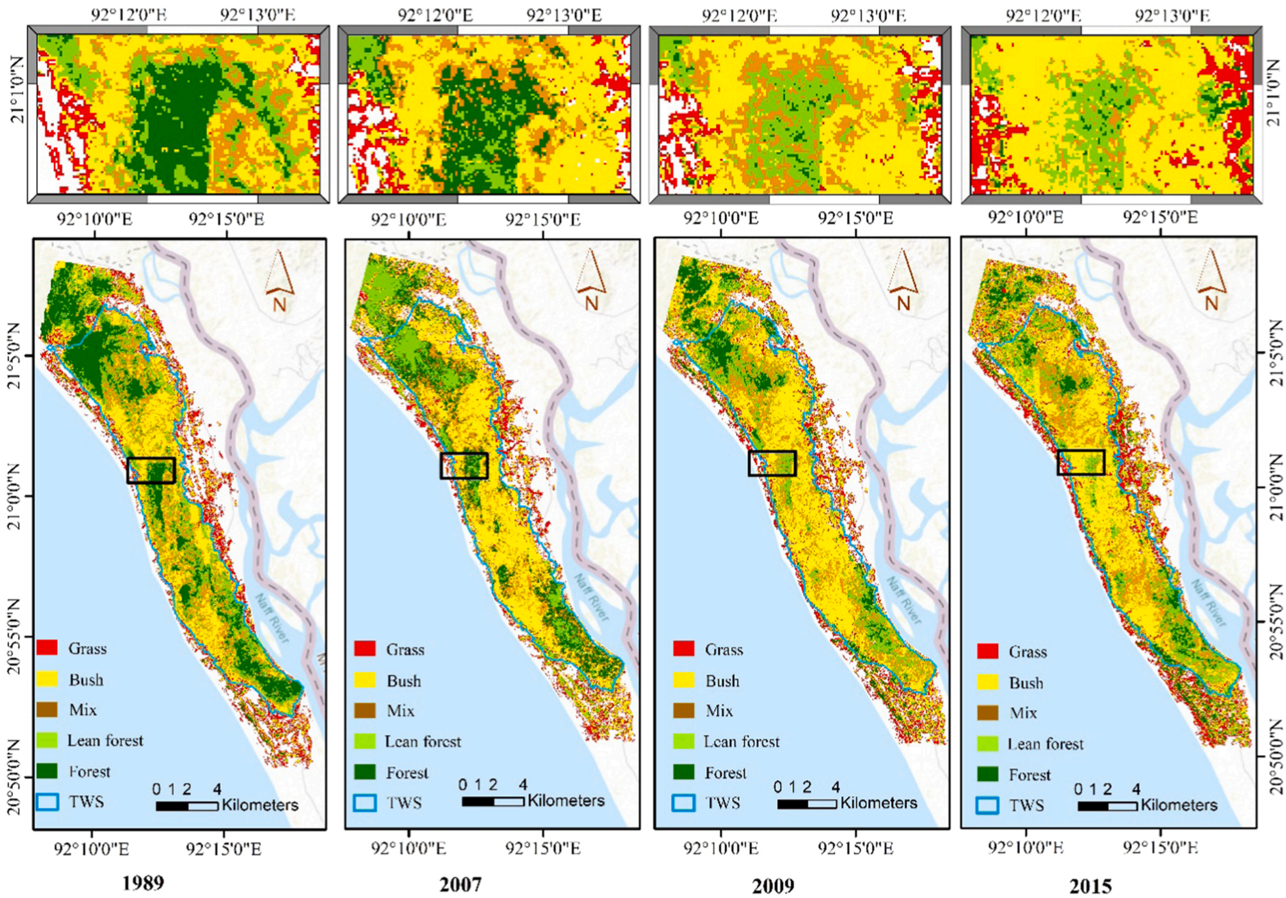

Fig. 4. Land-cover change of different land cover classes from 1989 to 2015 shown by raster images.

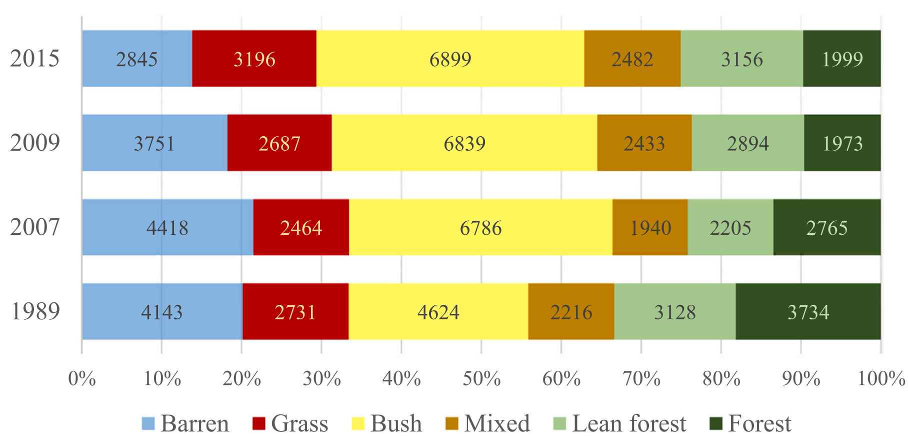

Fig. 5. Area of different land-cover classes regarding the change during 1989–2015.

constantly by losing 46% from 1989 to 2015. This pattern is a clear indication of forest destruction and conversion of forest lands into grass or bush in Teknaf Upazila.

4.2. Impact of TWS and co-management on the land-cover change and forest

TWS is a PA with a restricted boundary marking the forest area. The impact of TWS on the land-cover change and forests can be determined by the difference in the area change inside and outside of the TWS over the considered time. Table 2 shows the change in different land-cover classes inside and outside the TWS. As shown in Table 2, from 1989 to

Table 2
Land-cover class change inside and outside the TWS.

	Total		Inside TWS		Outside TWS	
	Area (ha)	%	Area (ha)	%	Area (ha)	%
Barren	-1298	-31	59	46	-1357	-34
Grass	465	17	251	56	213	9
Bush	2275	49	1621	50	654	47
Mixed	266	12	56	3	210	148
Lean forest	28	1	-286	-11	313	50
Forest	-1735	-46	-1701	-64	-34	-3

2015, forest decreased a total of 46%, but just inside the TWS, forest decreased 64%—much higher than just 3% outside the TWS. However, the changing pattern of grass followed the opposite trend to the forests. Grass increased several times more inside the TWS (56%) than the outside area (9%). Bush increased by approximately 50% both inside and outside of the TWS, but considering the land area expansion, bushes expanded 1621 ha inside, nearly three times that of the outside area (654 ha). Furthermore, bush had the largest land-cover class expansion for both inside and outside the TWS. Lean forest increased outside the TWS by 50% but decreased inside by 11%. Mixed changed a relatively small extent compared to the other land cover types.

Fig. 6 represents the comparison between the land-cover transformation process inside and outside the TWS. Inside the TWS, forests were gradually decreasing, whereas outside the TWS, forests decreased initially but began to increase after 2007. Lean forest followed a U-curve, where the initial decrease was followed by a gradual increase. Outside the TWS, lean forests increased from 626 ha to 988 ha between 1989 and 2007 and maintained coverage over 900 ha in 2015. Mixed remained comparatively stable but expanded outside the TWS. Bush and grass both expanded gradually inside and outside the TWS from 1989 to 2015. These land cover changes are also visible in Fig. 4, where the gradual decrease of dark green pixels representing the forests from the center area of the TWS is noticeable. Therefore, even inside the TWS, the forest land cover shrinks more than in the surrounding areas, and these forest areas are being replaced by bush and grasses. Moreover, lean forest areas are expanding outside the TWS, in contrast to the scenario occurring inside. These findings raise serious concerns over the effectiveness of the TWS in conserving forests.

A co-management approach was initiated in TWS in 2004, and since then, it has been followed with some adaptations and adjustments. The impact of co-management on forest conservation will be determined by comparing forest-cover change before and after the co-management implementation. Based on the available time-series data, 1989–2007 will be considered as before co-management implementation, and 2007–2015 will be considered as after co-management implementation.

Table 3 presents the forest-cover change before and after the co-management implementation. From 1989–2015, considering the total area, forest decreased by 1735 ha, among which 56% decreased before co-management, and 44% decreased after the co-management implementation. While considering only inside TWS, 1701 ha forest in total decreased from 1989 to 2015. Between that 1701 ha, 30% decreased before co-management implementation and 70% decreased after co-management implementation. Outside the TWS, 461 ha forest decreased before co-management, but after co-management implementation, forests increased by 427 ha, resulting in only a 3% total decrease during the period. It can be concluded that since the implementation of the co-management approach, forest loss has slowed in the region, but the co-management approach has failed to halt the deforestation inside TWS. Moreover, after the implementation of the co-management approach, deforestation increased inside the TWS, indicating a failure to protect the remaining natural forests.

Table 3
Comparing forest-cover change before and after co-management implementation.

	Total period		Before Co-management		After Co-management	
	Area	*% decrease*	*Area*	*% decrease of total period*	*Area*	*% decrease of total period*
Total	-1735	-46	-969	56	-766	44
TWS	-1701	-65	-508	-30	-1193	-70
Around	-34	-3	-461	–	+ 427	–

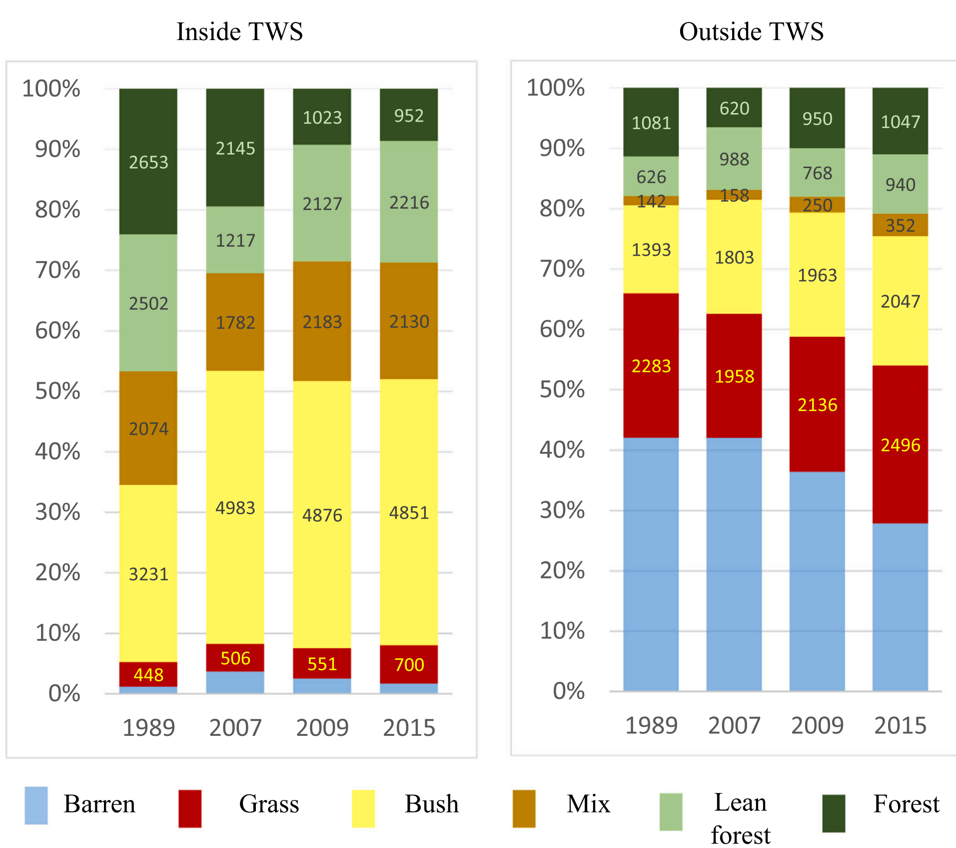

Fig. 6. Land-cover change comparison between inside and outside the TWS.

4.3. Impact of settlement on the land-cover change and forest

To understand the impact of settlement on forest land cover, the study area was divided into settlement and forest zones based on the vicinity of the households of the settlement. The land cover shifted from one type to another over the time period, but considering different zones, various patterns in this change indicated the impact of the settlements, as presented in Fig. 7.

In the forest zones (forest 1 km, forest 500 m, forest 200 m, and forest 100 m), forest land cover decreased by almost the same percentage, from 63% to 66%. Therefore, the impact of settlement on deforestation in the core areas of TWS is the same. However, in the case of settlement zones, the impact was different; from the core to the surrounding area (100 m settlement to 500 m settlement), except the 1 km settlement, the forest class gradually increased. Inside the core settlement zone (settlement 100 m), representing the closest surrounding area of the households, the forest class increased more than three times. For settlement at 200 m and 500 m, the increase in forest class was 178% and 20%, respectively. However, for the 1 km settlement zone, the forest class decreased by 33%. Considering the forest land class change among different zones, it is clear that in the forest zones, deforestation occurred at a constant rate, whereas moving toward the settlement zones, the closer to the households, the more the forests have increased.

For the lean forest, the change between different zones followed a similar pattern. Inside the forest zone, the lean forest decreased at a rate of 10–15%, and inside the settlement zones, it increased from 11% to 107%. The highest increase in lean forest (107%) occurred in the settlement 100 m zone. For the mixed land classes, all land classes increased among the different zones. For bush and grass, changes were opposite to the forest and lean forest land-cover change. Inside the forest zones, bush and grass increased more than in the settlement zones.

In particular, the increase of grass inside the forest zones was alarming. Inside the most core forest zones (i.e., forest 1 km zones), the grassland class increased almost 2 times (234%) from 1989 to 2015. Conversely, grassland class decreased by 12% in the core settlement areas (settlement 100 m) and had a lower rate of increase (9–15%) for settlement 500 m and settlement 1 km compared to forest zones. Thus, it is evident that forest decreased in the core areas of the TWS and increased around the settlement areas, resulting in the tree coverage shifting from the central forest areas toward the settlement areas. Moreover, the reduced forest land cover inside the TWS is being replaced by bush and grasses, resulting in an expansion of these land-cover categories inside the core of the TWS.

Fig. 7. Impact of settlement on the land-cover change by showing percentage of change between 1989 and 2015 of each land cover classes considering different settlement and forest zones.

5. Discussion

Declaring natural forests as PAs in various forms has been practiced in Bangladesh for more than a century. The concerned PA, TWS, has a record dating back to 1907 under the status of reserve forest. But the recent decrease in natural forests has created the need to assess the outcomes from these PAs in terms of conserving forests and their biodiversity. This study found that from 1989 to 2015, forest land cover decreased 46% in the Teknaf Peninsula and an alarming 64% just inside the PA referred to as TWS. Other studies using remote sensing to determine the land-cover change also detected forest-cover loss inside the TWS in recent times (Islam et al., 2020, 2019a, 2019b, 2019c). However, this study gave a better perspective of the impact of TWS on forest conservation by comparing the land-cover change inside and outside the TWS. Our work demonstrates that deforestation is occurring at a higher rate inside the TWS than outside, indicating that PAs are not yet attaining the desired outcome.

To stem deforestation inside PAs, co-management was introduced to govern the PAs in Bangladesh, and TWS has been under the co-management approach since 2004. The findings of the temporal comparison of the land-cover change before and after implementation of co-management (Table 3) show that after implementation, deforestation was occurring at a faster rate than before. Inside the TWS, 70% forest loss occurred after co-management implementation compared to just 30% before. The higher rate of forest loss indicates that the co-management is not yet successful in conserving the natural forests. Most recently, a study also found that co-management had concerning results conserving the forests in several other PAs in Bangladesh (Rahman and Islam, 2021). Co-management intends to involve local communities and provide sustainable benefits from forest resources while conserving the forests. Hence, to understand the outcome and impact of co-management approaches on conserving forests, it is important to determine the impact of the surrounding community, who are intended to be involved in forest management, on the forest-cover change.

Several other studies in Bangladesh have explored the forest-cover change of PAs by remote sensing (Ahmed and Bartlett, 2018; Billah et al., 2021; Chowdhury et al., 2020; Islam et al., 2020, 2019a, 2019b, 2019c; Rahman and Islam, 2021; Zafar et al., 2021). However, this study generated a better understanding of the process and nature of deforestation by exploring the impact of the surrounding settlements on the forest-cover change besides the temporal comparison of land covers inside the PAs. This study found that, besides shrinking forest cover inside the TWS, the tree coverage is shifting from the central reserve forest area toward the surrounding settlements. This observation indicates that the local people surrounding the TWS are cutting trees inside the TWS and, at the same time, planting trees at their homesteads. This process is more evident when considering the results showing that forests are being replaced by grass and bush inside the TWS, whereas tree coverage expanded significantly outside the PA boundary.

Besides the alarming rate of forest destruction in TWS, the shifting of greenness from the core reserved forest area toward its surroundings settlements raises some concerns. First, when planting trees near households, people typically plant exotic species trees for their benefits, including fruit, timber, and fodder. One study found that exotic tree species are dominant in some of the co-managed PAs in Bangladesh (Islam et al., 2020). Another reported the existence of a tree plantation near the TWS boundary (Asahiro, 2018). Planting exotic tree species is not an ideal solution when conserving natural forests, and planted trees cannot provide the same support for biodiversity as natural forests (Gibson et al., 2011). Second, most of the trees are planted near the forest boundary. These tree plantations require management and patrolling, which results in people more frequently entering the TWS. As the core forest area faces rapid deforestation, wild animals come to the forest boundaries more frequently for food and shelter. This trend makes the wild animals an easy target for traffickers and poachers and increases the conflict between animals and humans, which often results in the tragic death of wild animals. The conflict between wild animals and humans is already a concern for the PAs (Karanth et al., 2013; Nyhus, 2016), and action is required to mitigate this problem. However, in the case of TWS, the conflict between wildlife and humans may increase due to the shifting of trees from the core to the boundary, resulting in a further loss of biodiversity.

The continuous decrease of forest cover and the existence of large settlements surrounding the TWS have raised concerns over the effectiveness of the co-management approach. There are a handful of studies reporting the success of co-management in PAs to conserve forests and provide sustainable livelihood (Hajjar et al., 2021). To obtain such a positive impact from co-management in Bangladesh, two aspects must be considered: (1) strict and equal property rights and (2) capacity of the community to utilize those rights. In the case of TWS, this study found 5195 households inside the TWS, representing a legal dilemma. TWS is a restricted forest, but at the same time, there are households in approximately every 2 ha of area inside the forest boundary. TWS is supposed to restrict the entry of local people in order to preserve the natural forests, whereas a significant portion of the local community is living inside the restricted boundary. Most of these households are highly dependent on forest resources for their livelihood and have established homestead gardens with exotic tree species at the expense of cutting trees from natural forests. This “tragedy of the commons” scenario needs to be shifted towards “governing the commons,” and ensuring equal property rights is the prerequisite to do so. Establishing balanced and equal forest property rights between people living inside the TWS and people living outside the TWS is a major challenge for managing PAs like TWS. Property rights are widely understood as critical for sustainable land management and socioeconomic development in forest areas (Miller et al., 2021). Secure and equal property rights reduce the incentives of the local community to destroy forests and encourage them to invest in long-term improvement of the forest (Barbier and Tesfaw, 2013; Shyamsundar et al., 2020). Moreover, secure property rights can facilitate community control over natural resources to generate more income (Larson et al., 2012). However, co-management approaches cannot always ensure property rights, and there is evidence of reduced property rights after the implementation of the co-management approach (Hajjar et al., 2021). Therefore, it is crucial for the local community to have the self-capacity to utilize and protect property rights promised under the co-management. Capacity development of local communities reduces forest dependence and ensures effective participation in the management process (Devkota, 2020; Rashid and Khan, 2014). Positive conservation outcomes are more likely to occur when PAs lead to socioeconomic benefits for the local community (Oldekop et al., 2019b). Thus, empowering and benefitting the local community with secure forest property rights and implementing capacity development initiatives for the local community can play a crucial role in the success of the co-management approach in Bangladesh and conserving the remaining forests and their biodiversity.

The existing co-management model will continue to be followed in Bangladesh for the near future. This approach is embedded in the latest major policy changes, including the Forest Policy, 2016, and the Protected Area Management Rule, 2017, and is also included in future forestry plans such as the Bangladesh Forestry Master Plan 2017–2036. According to the existing model, the effective function of CMCs is the key to the successful conservation of forests. The BFD evaluated the CMCs under the CREL project in 2018 and found almost all CMCs to be satisfactory in terms of sustainability (Thompson et al., 2018). However, several field-based studies have addressed problems regarding the functionality and formation of CMCs, including power devolution, the dominance of government officials in decision-making, conflict with local elites, corruption, biased selection of beneficiaries, and inconsistency in the implementation of co-management approaches among the BFD officials (Belal, 2013; Chowdhury et al., 2014; Islam et al., 2019a, 2019b, 2019c; Manzoor Rashid and Khan, 2014; Uddin et al., 2019; Uddin and Parr, 2018). Moreover, CMCs receive funds to operate during the implementation period of DPA-funded projects and are supposed to be self-sufficient afterward through the income generated from forest resources they are bound to protect. Thus, CMCs can be a burden on PAs already facing deforestation. Budget shortages can seriously reduce the effectiveness of conserving biodiversity in PAs, and recently, declines in international aid have reduced the funding for PAs (Coad et al., 2019). Therefore, the DPA-funded project model of co-management will likely face serious challenges in the near future in Bangladesh.

For successful co-management in PAs, long-term land tenure for the local people in the forest boundary zone is crucial for shifting the dependency away from natural forests in the core area. However, the existing PA Rule (of 2017) has no scope for that. Villages within 5 km of the PA boundary are referred to as beneficiaries, but the BFD has no authority to lease land inside the PAs or adjacent areas of the forests to these beneficiaries (BFD, 2017). Even for leasing the areas adjacent to the forest, the district administration has the sole authority. A buffer zone, core zone, and wildlife corridors are mentioned in the PA rule 2017, but there is no clear instruction on how these zones should be determined and implemented. Another issue concerning forest policy is the resettlement policy framework under the SUFAL project. The draft of Bangladesh Forestry Master Plan 2017–2036 states that all the households inside PAs must be relocated outside (BFD and CRPARP, 2016). This kind of policy can accelerate conflict in and around the PA settlements, potentially hampering the conservation process. Resettlement initiatives can be implemented if there is a functioning buffer zone around the PAs where people can be relocated without completely changing their means of livelihood.

6. Conclusion

This study has shown that even after implementing co-management approaches, PAs in Bangladesh are losing forest covers. TWS, the PA considered for this study, experienced 64% forest loss inside its boundary and 46% forest loss in the surrounding region from 1989 to 2015. Besides forest-cover loss, the tree coverage is shifting from the central forest area toward the forest boundaries near the settlements. Even the homesteads surrounding the PA gained tree coverage during the measured period. Local communities tend to harvest forest resources from natural forest areas and manage homestead trees, which results in a shift of tree coverage from the core forest to the surroundings. Irreversible changes to the natural forest land cover in these PAs can lead to the rapid extinction of endangered species. Moreover, natural forests provide better ecosystem services, more biodiversity conservation, and an increased likelihood of reforestation of the surroundings (Estoque et al., 2019; Naughton-Treves et al., 2005c; Pan et al., 2011), which cannot be replicated by tree plantations. Therefore, prompt and strict actions are required to conserve the natural forest, and effective co-management approaches incorporating a long-term vision could be the answer.

Co-management is the latest governing and management tool for the PAs in Bangladesh with an aim to conserve the forests and their biodiversity and, at the same time, provide sustainable benefits to the surrounding communities. The success and impact of co-management are crucial for Bangladesh and require continuous monitoring and evolution initiatives. Besides co-management, the forest managers need to evaluate the actual status of the forests and predict the future transition for forest lands while developing forest management plans and strategies. In developing countries, forest transition is highly influenced by economic development and agricultural intensification (Singh et al., 2017; Wolfersberger et al., 2015), and both of these factors have changed rapidly in

Bangladesh during the past few years. The BFD must connect forest policies with forest transition trends and develop unique management plans for critical PAs such as TWS, where an appropriate balance should be maintained between "participation" and "protection" rather than simply following the prevailing trend to increase the tree coverage of the country through planting more trees.

CRediT authorship contribution Statement

SM Asik Ullah: Data curation and analysis, Conceptualization, Draft preparation, editing and reviewer response, finalizing draft. **Masakazu Tani:** Conceptualization, Funding acquisition, Draft review. **Jun Tsuchiya:** Data analysis and draft review. **M Abiar Rahman:** Data analysis and draft review. **Masao Moriyama:** Data analysis and draft review.

Acknowledgements

JSPS (Grant number: 15H02612, 19H00561 and 20K20020).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.landusepol.2021.105932.

References

Agarwal, B., 2001. Participatory exclusions, community forestry, and gender: an analysis for South Asia and a conceptual framework. World Dev. 29, 1623–1648. https://doi.org/10.1016/S0305-750X(01)00066-3.
Agrawal, A., 2007. Forests, governance, and sustainability: common property theory and its contributions. Int. J. Commons 1, 111. https://doi.org/10.18352/ijc.10.
Agrawal, A., Redford, K., 2009. Conservation and displacement: an overview. Conserv. Soc. 7, 1. https://doi.org/10.4103/0972-4923.54790.
Ahmed, S., Bartlett, D., 2018. An evaluation of the effectiveness of the co-management approach in selected protected areas of Bangladesh. Int. J. Biodivers. Conserv. 10, 510–516. https://doi.org/10.5897/IJBC2017.1107.
Alam, M.F., Uddin, M.Z., Hasan, M.A., 2013. Evaluation of Plant Biodiversity in Teknaf Wildlife Sanctuary, B.desh Evaluation, Plant, Biodiversity, Teknaf Wildlife Sanctuary. LAP LAMBERT Academic Publishing, Bangladesh.
Asahiro, K., 2018. Social forestry conditions on the Teknaf Peninsula. In: Deforestation in the Teknaf Peninsula of Bangladesh. Springer, Singapore, Singapore, pp. 129–142. https://doi.org/10.1007/978-981-10-5475-4_10.
Aziz, N., Decosee, P.J., 2009. Nishorgo Support Project Completion Report.
Balmford, A., Gaston, K.J., Blyth, S., James, A., Kapos, V., 2003. Global variation in terrestrial conservation costs, conservation benefits, and unmet conservation needs. Proc. Natl. Acad. Sci. U.S.A 100, 1046–1050. https://doi.org/10.1073/pnas.0236945100.
Barbier, E.B., Tesfaw, A.T., 2013. Tenure constraints and carbon forestry in Africa. Am. J. Agric. Econ. 95, 964–975. https://doi.org/10.1093/AJAE/AAT014.
BBS, 2011. Population-and-Housing-Census - Bangladesh Bureau of Statistics-Government of the People\'s Republic of Bangladesh [WWW Document]. URL ⟨http://www.bbs.gov.bd/site/page/47856ad0–7e1c-4aab-bd78–892733bc06eb/Population-and-Housing-Census⟩ (Accessed 9 August 2021).
Belal, M.A.H., 2013. Effectiveness of Co-management Committees in Teknaf Wildlife Sanctuary, in: Jefferson, F., M.G., M., Bryan, R Bushley, Stephen, M.B., Laurie, D. (Eds.), CONNECTING COMMUNITIES AND CONSERVATION: CO-MANAGEMENT INITIATIVES IMPLEMENTED BY IPAC IN WETLANDS AND FORESTS OF BANGLADESH. Dhaka: USAID; Integrated Protected Area Co-Management Project, pp. 10–23.
BFD, 2021. Bangladesh Forest Department [WWW Document]. URL ⟨http://www.bforest.gov.bd/⟩ (accessed 12.6.21).
BFD, 2020. Bangladesh Forest Department - Protected Area [WWW Document]. ⟨https://bforest.portal.gov.bd/site/page/5430ce33–561e-44f6–9827-ea1ebaa2c00d/-⟩ (Accessed 11 March 2020).
BFD, 2017. Bangladesh Forest Department - Protected Area Co-management Rule,2017 [WWW Document]. URL ⟨https://bforest.portal.gov.bd/sites/default/files/files/bforest.portal.gov.bd/page/a2f633e5_8b6c_4213_b78c_ec966bd2a942/PA⟩ Rule 2017. pdf (Accessed 8 June 20).
BFD, 2016. National Forest Policy-2016, Policy-Bangladesh Forest Department.
BFD, 2012. Bangladesh Forest Department - Wildlife (Conservation and Security) Act, 2012 [WWW Document]. URL ⟨https://bforest.portal.gov.bd/sites/default/files/files/bforest.portal.gov.bd/page/bcc3e216_3484_4e19_b2fa_5e771b3c3557/Wildlife⟩ Act 2012.pdf (Accessed 18 June 2020).
BFD, 2004. Bangladesh Forest Department - Social Forestry Rule, 2004.
BFD, 1994. Bangladesh Forest Department - National Forest Policy,1994.
BFD, CRPARP, 2016. Bangladesh forestry master plan 2017–2036.
Billah, M.M., Rahman, M.M., Abedin, J., Akter, H., 2021. Land cover change and its impact on human–elephant conflict: a case from Fashiakhali forest reserve in Bangladesh. SN Appl. Sci. 36 3, 1–17. https://doi.org/10.1007/S42452-021-04625-1.
Brockington, D., Igoe, J., Schmidt-Soltau, K., 2006. Conservation, human rights, and poverty reduction. Conserv. Biol. https://doi.org/10.1111/j.1523-1739.2006.00335.x.
Bruner, A.G., Gullison, R.E., Balmford, A., 2004. Financial costs and shortfalls of managing and expanding protected-area systems in developing countries. Bioscience 54, 1119–1126. https://doi.org/10.1641/0006-3568(2004)054[1119:fcasom]2.0.co;2.
Butchart, S.H.M., Scharlemann, J.P.W., Evans, M.I., Quader, S., Aricò, S., Arinaitwe, J., Balman, M., Bennun, L.A., Bertzky, B., Besançon, C., Boucher, T.M., Brooks, T.M., Burfield, I.J., Burgess, N.D., Chan, S., Clay, R.P., Crosby, M.J., Davidson, N.C., De Silva, N., Devenish, C., Dutson, G.C.L., Fernández, D.F.D. z, Fishpool, L.D.C., Fitzgerald, C., Foster, M., Heath, M.F., Hockings, M., Hoffmann, M., Knox, D., Larsen, F.W., Lamoreux, J.F., Loucks, C., May, I., Millett, J., Molloy, D., Morling, P., Parr, M., Ricketts, T.H., Seddon, N., Skolnik, B., Stuart, S.N., Upgren, A., Woodley, S., 2012. Protecting important sites for biodiversity contributes to meeting global conservation targets. PLoS One 7, e32529. https://doi.org/10.1371/journal.pone.0032529.
Chowdhury, F.I., Islam, K., Faroque, M., Al, Islam, K.N., Rahman, M.F., Arif, M.T.A., Nath, T.K., Jashimuddin, M., 2020. Assessing the impacts of co-management on protected area landscape under socio-imagery lens: evidence from Bangladesh. J. Sustain. For. https://doi.org/10.1080/10549811.2020.1747497 https://doi.org/10.1080/10549811.2020.1747497.
Chowdhury, M.S.H., Koike, M., 2010. An overview on the protected area system for forest conservation in Bangladesh. J. . Res. 21, 111–118. https://doi.org/10.1007/s11676-010-0019-x.
Chowdhury, M.S.H., Nazia, N., Izumiyama, S., Muhammed, N., Koike, M., 2014. Patterns and extent of threats to the protected areas of Bangladesh: the need for a relook at conservation strategies. PARKS 20, 91–104. https://doi.org/10.2305/IUCN.CH.2014.PARKS-20-1.MSHC.en.
Coad, L., Watson, J.E., Geldmann, J., Burgess, N.D., Leverington, F., Hockings, M., Knights, K., Di Marco, M., 2019. Widespread shortfalls in protected area resourcing undermine efforts to conserve biodiversity. Front. Ecol. Environ. 17, 259–264. https://doi.org/10.1002/fee.2042.
Cox, M., Arnold, G., Tomás, S.V., 2010. A review of design principles for community-based natural resource management. Ecol. Soc. https://doi.org/10.5751/ES-03704-150438.
Devkota, B.P., 2020. Social inclusion and deliberation in response to REDD+ in Nepal's community forestry. . Policy Econ. 111, 102048 https://doi.org/10.1016/j.forpol.2019.102048.
Estoque, R.C., Ooba, M., Avitabile, V., Hijioka, Y., DasGupta, R., Togawa, T., Murayama, Y., 2019. The future of Southeast Asia's forests. Nat. Commun. 10, 1829. https://doi.org/10.1038/s41467-019-09646-4.
FAO, 2020. Global Forest Resources Assessment 2020: Main Report, Global Forest Resources Assessment 2020. FAO, Rome. https://doi.org/10.4060/ca8753en.
FAO, 2016. Global Forest Resources Assessments 2015. How are the World's Forest Changing? Food and Agriculture Organization of the United Nations, Rome.
FAO, 2015. Global Forest Resources Assesment. Rome.
Geist, H.J., Lambin, E.F., 2002. Proximate causes and underlying driving forces of tropical deforestation. Bioscience 52, 143–150 https://doi.org/10.1641/0006-3568(2002)052[0143:PCAUDF]2.0.CO;2.
Geldmann, J., Barnes, M., Coad, L., Craigie, I.D., Hockings, M., Burgess, N.D., 2013a. Systematic review effectiveness of terrestrial protected areas in reducing habitat loss and population declines. Biol. Conserv. 161, 230–238. https://doi.org/10.1016/j.biocon.2013.02.018.
Geldmann, J., Barnes, M., Coad, L., Craigie, I.D., Hockings, M., Burgess, N.D., 2013b. Effectiveness of terrestrial protected areas in reducing habitat loss and population declines. Biol. Conserv. https://doi.org/10.1016/j.biocon.2013.02.018.
Gibson, L., Lee, T.M., Koh, L.P., Brook, B.W., Gardner, T.A., Barlow, J., Peres, C.A., Bradshaw, C.J.A., Laurance, W.F., Lovejoy, T.E., Sodhi, N.S., 2011. Primary forests are irreplaceable for sustaining tropical biodiversity. Nature 478, 378–381. https://doi.org/10.1038/nature10425.
Hajjar, R., Oldekop, J.A., Cronkleton, P., Newton, P., Russell, A.J.M., Zhou, W., 2021. A global analysis of the social and environmental outcomes of community forests. Nat. Sustain. 4, 216–224. https://doi.org/10.1038/s41893-020-00633-y.
Hajjar, R., Oldekop, J.A., Cronkleton, P., Newton, P., Russell, A.J.M., Zhou, W., 2020. A global analysis of the social and environmental outcomes of community forests, 2020 Nat. Sustain. 43 4, 216–224. https://doi.org/10.1038/s41893-020-00633-y.
Hardin, G., 1968. The tragedy of the commons. The population problem has no technical solution; it requires a fundamental extension in morality. Science 162, 1243–1248. https://doi.org/10.1126/SCIENCE.162.3859.1243.
Islam, K., Nath, T.K., Jashimuddin, M., Rahman, M.F., 2019a. Forest dependency, co-management and improvement of peoples' livelihood capital: Evidence from Chunati Wildlife Sanctuary. Bangladesh Environ. Dev. 32, 100456 https://doi.org/10.1016/J.ENVDEV.2019.100456.
Islam, K., Nath, T.K., Jashimuddin, M., Rahman, M.F., 2019b. Forest dependency, co-management and improvement of peoples' livelihood capital: evidence from Chunati Wildlife Sanctuary. Bangladesh Environ. Dev. 32, 100456 https://doi.org/10.1016/J.ENVDEV.2019.100456.
Islam, K.N., Rahman, M.M., Jashimuddin, M., Hossain, M.M., Islam, K., Faroque, M.Al, 2019c. Analyzing multi-temporal satellite imagery and stakeholders' perceptions to have an insight into how forest co-management is changing the protected area landscapes in Bangladesh. . Policy Econ. 101, 70–80. https://doi.org/10.1016/J.FORPOL.2019.01.011.

Islam, K.N., Rahman, M.M., Jashimuddin, M., Islam, K., Zhang, Y., 2020. Impact of co-management on tree diversity and carbon sequestration in protected areas: experiences from Bangladesh. Trees, . People 2, 100033. https://doi.org/10.1016/J.TFP.2020.100033.

Joppa, L., Pfaff, A., 2010. Reassessing the forest impacts of protection the challenge of nonrandom location and a corrective method. Ann. N. Y. Acad. Sci.

Karanth, K.K., Naughton-Treves, L., Defries, R., Gopalaswamy, A.M., 2013. Living with wildlife and mitigating conflicts around three indian protected areas. Environ. Manag. 52, 1320–1332. https://doi.org/10.1007/s00267-013-0162-1.

Larson, A.M., Barry, D., Dahal, G.R., Pierce Colfer, C.J., 2012. Forests for people: Community rights and forest tenure reform, Forests for People: Community Rights and Forest Tenure Reform. Taylor and Francis Group,, Routledge.

Lemos, M.C., Agrawal, A., 2006. Environmental Governance. Annu. Rev. Environ. Resour. 31, 297–325. https://doi.org/10.1146/annurev.energy.31.042605.135621.

Leverington, F., Costa, K.L., Pavese, H., Lisle, A., Hockings, M., 2010. A global analysis of protected area management effectiveness. Environ. Manag. 46, 685–698. https://doi.org/10.1007/s00267-010-9564-5.

Mackenzie, C., Etter, L., Chowdhury, A.J.M.I., Islam, S., 2013. Performance Evaluation of the Integrated Protected Areas Co-management (IPAC) Project | Global Climate Change. Dhaka.

Manzoor Rashid, A.Z.M., Khan, N.A., 2014. Role of Co-management Organizations in Protected Area Governance: Some Observations from the Chunati Wildlife Sanctuary. Springer,, Cham, pp. 181–200. https://doi.org/10.1007/978-3-319-08147-2_9.

Matose, F., 2006. Co-management options for reserved forests in Zimbabwe and beyond: policy implications of forest management strategies. . Policy Econ. 8, 363–374. https://doi.org/10.1016/J.FORPOL.2005.08.013.

Miller, D.C., Rana, P., Nakamura, K., Irwin, S., Cheng, S.H., Ahlroth, S., Perge, E., 2021. A global review of the impact of forest property rights interventions on poverty. Glob. Environ. Chang. 66, 102218 https://doi.org/10.1016/j.gloenvcha.2020.102218.

Mollick, S.A., Khalilur Rahman, M., Nabiul Islam Khan, M., Nazmus Sadath, M., 2018. Evaluation of good governance in a participatory forestry program: A case study in Madhupur Sal forests of Bangladesh. . Policy Econ. 95, 123–137. https://doi.org/10.1016/J.FORPOL.2018.07.014.

Moslehuddin, A.Z.M., Rahman, M.A., Ullah, S.M.A., Moriyama, M., Tani, M., 2018. Physiography, Forests, and People in Teknaf. In: Deforestation in the Teknaf Peninsula of Bangladesh. Springer Singapore, Singapore, pp. 11–40. https://doi.org/10.1007/978-981-10-5475-4_.

Muench, C., Martínez-Ramos, M., 2016. Can community-protected areas conserve biodiversity in human-modified tropical landscapes? The case of terrestrial mammals in Southern Mexico. Trop. Conserv. Sci. 9, 178–202. https://doi.org/10.1177/194008291600900110.

Muhammed, N., Koike, M., Haque, F., Miah, M.D., 2008. Quantitative assessment of people-oriented forestry in Bangladesh: a case study in the Tangail forest division. J. Environ. Manag. 88, 83–92. https://doi.org/10.1016/J.JENVMAN.2007.01.029.

Mukul, S.A., Rashid, A.Z.M.M., Quazi, S.A., Uddin, M.B., Fox, J., 2012. Local peoples' responses to co-management regime in protected areas: A case study from Satchari National Park, Bangladesh. For. Trees Livelihoods 21, 16–29. https://doi.org/10.1080/14728028.2012.669132.

Naughton-Treves, L., Holland, M.B., Brandon, K., 2005a. The role of protected areas in conserving biodiversity and sustaining local livelihoods. https://doi.org/10.1146/annurev.energy.30.050504.164507.

Naughton-Treves, L., Holland, M.B., Brandon, K., 2005b. The role of protected areas in conserving biodiversity and sustaining local livelihoods. Annu. Rev. Environ. Resour. 30, 219–252. https://doi.org/10.1146/annurev.energy.30.050504.164507.

Naughton-Treves, L., Holland, M.B., Brandon, K., 2005c. The role of protected areas in conserving biodiversity and sustaining local livelihoods. Annu. Rev. Environ. Resour. 30, 219–252. https://doi.org/10.1146/annurev.energy.30.050504.164507.

Nyhus, P.J., 2016. Human-Wildlife Conflict and Coexistence. https://doi.org/10.1146/annurev-environ-110615–085634.

Oldekop, J.A., Sims, K.R.E., Karna, B.K., Whittingham, M.J., Agrawal, A., 2019a. Reductions in deforestation and poverty from decentralized forest management in Nepal. Nat. Sustain. 25 2, 421–428. https://doi.org/10.1038/s41893-019-0277-3.

Oldekop, J.A., Sims, K.R.E., Karna, B.K., Whittingham, M.J., Agrawal, A., 2019b. Reductions in deforestation and poverty from decentralized forest management in Nepal. Nat. Sustain. 2, 421–428. https://doi.org/10.1038/s41893-019-0277-3.

Ostrom, Elinor, 1990a. Governing the Commons, Governing the Commons. Cambridge University Press. https://doi.org/10.1017/cbo9780511807763.

Ostrom, Elinor, 1990b. Governing the Commons: The Evolution of Institutions For Collective Action. Cambridge University Press.

Pagdee, A., Kim, Y.S., Daugherty, P.J., 2006. What makes community forest management successful: a meta-study from community forests throughout the world. Soc. Nat. Resour. 19, 33–52 https://doi.org/10.1080/08941920500323260.

Pan, Y., Birdsey, R.A., Fang, J., Houghton, R., Kauppi, P.E., Kurz, W.A., Phillips, O.L., Shvidenko, A., Lewis, S.L., Canadell, J.G., Ciais, P., Jackson, R.B., Pacala, S.W., McGuire, A.D., Piao, S., Rautiainen, A., Sitch, S., Hayes, D., 2011. A large and persistent carbon sink in the World's forests. Science 333, 988–993. https://doi.org/10.1126/science.1201609.

Peluso, N.L., 2012. Rich Forests, Poor PeopleResource Control and Resistance in Java, Rich Forests, Poor PeopleResource Control and Resistance in Java. University of California Press. https://doi.org/10.1525/california/9780520073777.001.0001.

Pfaff, A., Amacher, G.S., Sills, E.O., 2013. Realistic REDD: improving the forest impacts of domestic policies in different settings. Rev. Environ. Econ. Policy 7, 114–135. https://doi.org/10.1093/reep/res023.

Potapov, P., Siddiqui, B.N., Iqbal, Z., Aziz, T., Zzaman, B., Islam, A., Pickens, A., Talero, Y., Tyukavina, A., Turubanova, S., Hansen, M.C., 2017. Comprehensive monitoring of Bangladesh tree cover inside and outside of forests, 2000 - 2014. Environ. Res. Lett. 12, 104015 https://doi.org/10.1088/1748-9326/aa84bb.

Pringle, R.M., 2017. Upgrading protected areas to conserve wild biodiversity. Nature. https://doi.org/10.1038/nature22902.

Rahman, M.F., Islam, K., 2021. Effectiveness of protected areas in reducing deforestation and forest fragmentation in Bangladesh. J. Environ. Manag. 280, 111711 https://doi.org/10.1016/J.JENVMAN.2020.111711.

Rahman, M.H., Miah, M.D., Rahman, M.H., Miah, M.D., 2017. Are protected forests of bangladesh prepared for the implementation of REDD+? A forest governance analysis from rema-kalenga wildlife sanctuary. Environments 4, 43. https://doi.org/10.3390/environments4020043.

Rahman, M.S., Sadath, M.N., Giessen, L., 2016a. Foreign donors driving policy change in recipient countries: three decades of development aid towards community-based forest policy in Bangladesh. For. Policy Econ. 68, 39–53. https://doi.org/10.1016/J.FORPOL.2016.03.013.

Rahman, M.S., Sarker, P.K., Giessen, L., 2016b. Power players in biodiversity policy: insights from international and domestic forest biodiversity initiatives in Bangladesh from 1992 to 2013. Land Use Policy 59, 386–401. https://doi.org/10.1016/J.LANDUSEPOL.2016.09.013.

Rashid, A.Z.M., Craig, D., Jeffery, M.I., Khan, N.A., 2013a. Forest protected area governance in Bangladesh: a focus on the legal and policy framework. Chin. J. Popul. Resour. Environ. 11, 345–351. https://doi.org/10.1080/10042857.2013.868576.

Rashid, A.Z.M., Craig, D., Mukul, S.A., Khan, N.A., 2013b. A journey towards shared governance: status and prospects for collaborative management in the protected areas of Bangladesh. J. . Res. 24, 599–605. https://doi.org/10.1007/s11676-013-0391-4.

RRI, 2017. Forest and Land Tenure | Rights + Resources.

Rudel, T.K., Defries, R., Asner, G.P., Laurance, W.F., 2009. Changing drivers of deforestation and new opportunities for conservation. Conserv. Biol. 23, 1396–1405. https://doi.org/10.1111/j.1523-1739.2009.01332.x.

S Alam, K.M.M.R.M.K., 2015. Threats to the Teknaf wildlife sanctuary of Bangladesh. J. Environ. Sci. Nat. Resour. 7, 233–239. https://doi.org/10.3329/jesnr.v7i1.22176.

Sadath, M.N., Krott, M., 2012. Identifying policy change — analytical program analysis: an example of two decades of forest policy in Bangladesh. . Policy Econ. 25, 93–99. https://doi.org/10.1016/J.FORPOL.2012.07.013.

Shyamsundar, P., Ahlroth, S., Kristjanson, P., Onder, S., 2020. Supporting pathways to prosperity in forest landscapes – a PRIME framework. World Dev. 125, 104622 https://doi.org/10.1016/J.WORLDDEV.2019.104622.

Singh, M.P., Bhojvaid, P.P., Ashraf, J., Reddy, S.R., 2017. Forest transition and socio-economic development in India and their implications for forest transition theory. Policy Econ. 76, 65–71. https://doi.org/10.1016/J.FORPOL.2015.10.013.

Tani, M., 2018a. Forest dwellers and encroachers in teknaf. In: Tani, M., Abiar, M.R. (Eds.), Deforestation in the Teknaf Peninsula of Bangladesh: A Study of Political Ecology. Springer, Singapore, pp. 101–112. https://doi.org/10.1007/978-981-10-5475-4_8.

Tani, M., Rahman, M.A., 2018b. Introduction. In: Tani, M., Rahman, M.A. (Eds.), Deforestation in the Teknaf Peninsula of Bangladesh: A Study of Political Ecology. Springer, Singapore, pp. 3–10. https://doi.org/10.1007/978-981-10-5475-4_1.

Thompson, P., Chowdhury, S.N., 2007. Experiences in wetland co-management: the MACH project | WorldFish Publications. In: Dickson, M., Brooks, A. (Eds.), Proceedings of the CBFM-2 International Conference on Community Based Approaches to Fisheries Management. The WorldFish Center - Bangladesh Office, Dhaka.

Thompson, P.M., Chowdhury, R.M., Uddin, M.S., Dutta, U., 2018. Capacity and Sustainability of Co-Management Organizations. CREL Technical Report No. 4. Climate-Resilient Ecosystems and Livelihoods (CREL) Project. Dhaka, Bangladesh.

Tucker, C.J., 1979. Red and photographic infrared linear combinations for monitoring vegetation. Remote Sens. Environ. 8, 127–150. https://doi.org/10.1016/0034-4257(79)90013-0.

Uddin, M.N., Hossain, M.M., Chen, Y., Siriwong, W., Boonyanuphap, J., 2019. Stakeholders' perception on indigenous community-based management of village common forests in Chittagong hill tracts, Bangladesh. . Policy Econ. 100, 102–112. https://doi.org/10.1016/J.FORPOL.2018.12.005.

Uddin, S., Parr, J.W.K., 2018. Evaluating collaborative management within the national parks and wildlife sanctuaries of bangladesh. Parks 24, 51–66. https://doi.org/10.2305/IUCN.CH.2018.PARKS-24-1MSU.en.

Ullah, A.S.M., Tani, M., Tsuchiya, J., Rahman, A.M., Rahman, Z.M., 2020. Impact of Betel Leaf Cultivation on the protected forest area of Teknaf Peninsula, Bangladesh. Small Scale For. 193 19, 335–355. https://doi.org/10.1007/S11842-020-09441-W.

Ullah, S.M.A., Tsuchiya, J., 2018. Fuelwood of teknaf forests: marketing, distribution, and concomitant people. In: Tani, M., Rahman, M.A. (Eds.), Deforestation in the Teknaf Peninsula of Bangladesh: A Study of Political Ecology. Springer, Singapore, pp. 85–100. https://doi.org/10.1007/978-981-10-5475-4_7.

UNEP-WCMC, 2020. Explore the World's Marine Protected Areas.

USAID, 2018. Conserving Ecosystems to Support Climate Resilience in Bangladesh- An Ecosystem-based Adaptation Approach. Washington, DC 20523.

Venter, O., Fuller, R.A., Segan, D.B., Carwardine, J., Brooks, T., Butchart, S.H.M., Di Marco, M., Iwamura, T., Joseph, L., O'Grady, D., Possingham, H.P., Rondinini, C., Smith, R.J., Venter, M., Watson, J.E.M., 2014. Targeting global protected area expansion for imperiled biodiversity. PLoS Biol. 12, e1001891 https://doi.org/10.1371/journal.pbio.1001891.

Watson, J.E.M., Darling, E.S., Venter, O., Maron, M., Walston, J., Possingham, H.P., Dudley, N., Hockings, M., Barnes, M., Brooks, T.M., 2016. Bolder science needed

now for protected areas. Conserv. Biol. 30, 243–248. https://doi.org/10.1111/cobi.12645.
Watson, James, E.M., Dudley, N., Segan, D.B., Hockings, M., 2014a. The performance and potential of protected areas. Nature. https://doi.org/10.1038/nature13947.
Watson, James, E.M., Dudley, N., Segan, D.B., Hockings, M., 2014b. The performance and potential of protected areas. Nature 67—73. https://doi.org/10.1038/nature13947.
Wolfersberger, J., Delacote, P., Garcia, S., 2015. An empirical analysis of forest transition and land-use change in developing countries. Ecol. Econ. 119, 241–251. https://doi.org/10.1016/J.ECOLECON.2015.08.018.
World Bank, 2018. Bangladesh - Sustainable Forests and Livelihood (SUFAL) Project: resettlement plan: Resettlement policy framework (English). Bangladesh.
Yang, H., Viña, A., Winkler, J.A., Chung, M.G., Huang, Q., Dou, Y., McShea, W.J., Songer, M., Zhang, J., Liu, J., 2021. A global assessment of the impact of individual protected areas on preventing forest loss. Sci. Total Environ. 777, 145995 https://doi.org/10.1016/J.SCITOTENV.2021.145995.
Zafar, T.B., Ding, W., Din, S.U., Khan, G.M., Hao, C., He, L., 2021. Forest cover and land use map of the Chunati Wildlife Sanctuary based on participatory mapping and satellite images: insight into Chunati beat. Land Use Policy 103, 105193. https://doi.org/10.1016/J.LANDUSEPOL.2020.105193.